Cuadro 1: Pruebas relacionadas con medias

H_0	**Valor del estadístico de prueba**	H_1	**Región crítica**
$\mu = \mu_0$	$z = \dfrac{\bar{x} - \mu_0}{\sigma/\sqrt{n}},\ \sigma$ conocida	$\mu < \mu_0$	$z < -z_\alpha$
		$\mu > \mu_0$	$z > z_\alpha$
		$\mu \neq \mu_0$	$\lvert z\rvert > z_{\alpha/2}$
$\mu = \mu_0$	$t = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}},\ v = n-1$	$\mu < \mu_0$	$t < -t_\alpha$
		$\mu > \mu_0$	$t > t_\alpha$
		$\mu \neq \mu_0$	$\lvert t\rvert > t_{\alpha/2}$
$\mu_1 - \mu_2 = d_0$ σ_1, σ_2 conocidas	$z = \dfrac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}}$	$\mu_1 - \mu_2 < d_0$	$z < -z_\alpha$
		$\mu_1 - \mu_2 > d_0$	$z > z_\alpha$
		$\mu_1 - \mu_2 \neq d_0$	$\lvert z\rvert > z_{\alpha/2}$
$\mu_1 - \mu_2 = d_0$ $\sigma_1 = \sigma_2$ desconocida	$t = \dfrac{(\bar{x}_1 - \bar{x}_2) - d_0}{s_p\sqrt{1/n_1 + 1/n_2}}$ $s_p^2 = \dfrac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}$ $v = n_1 + n_2 - 2$	$\mu_1 - \mu_2 < d_0$	$t < -t_\alpha$
		$\mu_1 - \mu_2 > d_0$	$t > t_\alpha$
		$\mu_1 - \mu_2 \neq d_0$	$\lvert t\rvert > t_{\alpha/2}$
$\mu_1 - \mu_2 = d_0$ $\sigma_1 \neq \sigma_2$ desconocida	$t' = \dfrac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}$ $v \approx \dfrac{(s_1^2/n_1 + s_2^2/n_2)^2}{(s_1^2/n_1)^2/(n_1-1) + (s_2^2/n_2)^2/(n_2-1)}$	$\mu_1 - \mu_2 < d_0$	$t' < -t_{\alpha/2}$
		$\mu_1 - \mu_2 > d_0$	$t' > t_{\alpha/2}$
		$\mu_1 - \mu_2 \neq d_0$	$\lvert t'\rvert > t_{\alpha/2}$
$\mu_D = d_0$ Observaciones pareadas	$t = \dfrac{d - d_0}{s_d/\sqrt{n}},\ v = n-1$	$\mu_D < d_0$	$t < -t_\alpha$
		$\mu_D > d_0$	$t > t_\alpha$
		$\mu_D \neq d_0$	$\lvert t\rvert > t_{\alpha/2}$
$p = p_0$	$z = \dfrac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}$	$p < p_0$	$z < -z_\alpha$
		$p > p_0$	$z > z_\alpha$
		$p \neq p_0$	$\lvert z\rvert > z_{\alpha/2}$
$p_1 - p_2 = 0$ $\hat{p} = \dfrac{x_1 + x_2}{n_1 + n_2}$	$z = \dfrac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$	$p_1 - p_2 < 0$	$z < -z_\alpha$
		$p_1 - p_2 > 0$	$z > z_\alpha$
		$p_1 - p_2 \neq 0$	$\lvert z\rvert > z_{\alpha/2}$
$\sigma^2 = \sigma_0^2$	$\chi^2 = \dfrac{(n-1)s^2}{\sigma_0^2},\ v = n-1$	$\sigma^2 < \sigma_0^2$	$\chi^2 < \chi^2_\alpha$
		$\sigma^2 > \sigma_0^2$	$\chi^2 > \chi^2_{1-\alpha}$
		$\sigma^2 \neq \sigma_0^2$	$\chi^2 < \chi^2_{\alpha/2}$ o $\chi^2 > \chi^2_{1-\alpha/2}$
$\sigma_1^2 = \sigma_2^2$ $v_1 = n_1 - 1,\ v_2 = n_2 - 1$	$F = \dfrac{s_1^2}{s_2^2}$	$\sigma_1^2 < \sigma_2^2$	$F < F_\alpha$
		$\sigma_1^2 > \sigma_2^2$	$F > F_{1-\alpha}$
		$\sigma_1^2 \neq \sigma_2^2$	$F < F_{\alpha/2}$ o $F > F_{1-\alpha/2}$